

Weekly CT-01

Cambridge International AS and A Level Computer Science 9618

Data Representation Multimedia Representation and Compression

Questions: 6

Total Marks: 20

Answer all six questions. Show all working for calculations.

1. A scanned image is a bitmap image. Describe each term.

[2]

(a) Pixel

(b) File header

2. An image has a resolution of 1024 x 512 pixels and a colour depth of 8 bits per pixel. Calculate an estimate of the file size in mebibytes. Show your working.

[3]

3. Identify one lossless compression method that can be used to compress a bitmap image. Describe how the method reduces the file size.

[2]

Method: _____

4. A computer uses the ASCII character set.

[4]

(a) State the number of characters represented by ASCII and extended ASCII. [2]

ASCII: _____ Extended ASCII: _____

(b) Explain how a word such as HOUSE is represented using ASCII. [2]

5. A sound is recorded several times using a different sample rate each time. [4]

(a) Explain why the recorded sound is closer to the original when a higher sample rate is used. [2]

(b) Explain why the sound file size increases when a higher sample rate is used. [2]

6. A student compresses a sound file before sending it by email.

[4]

(a) Explain why the student compresses the sound file. [2]

(b) Describe how lossless compression can compress the sound file. [2]

7. How long did you study for this short exam ?

[0.5]

Hour = _____

8. Thank you for participating the exam honestly.

[0,5]